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Inheritance

Learning Objectives

After the completion of this chapter, the student will be able to

- Understand the purpose of Inheritance
- Construct C++ programs using Inheritance
- Execute and debug programs which contains the concept of Inheritance

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16.1 Introduction to Inheritance

Inheritance is one of the most important features of Object Oriented Programming is Inheritance.In object-oriented programming, inheritance enables new class and its objects to take on the properties of the existing classes. A class that is used as the basis forinheritance is called a superclass or base class. A class that inherits from a superclass is called a subclass or derived class

16.2 Need for Inheritance

Inheritance is an important feature of object oriented programming used for code reusability. It is a process of creating new classes called derived classes, from the existing or base classes. Inheritance allows us to inherit all the code (except declared as private) of one class to another class. The class to be inherited is called base class or parent class and the class which inherits the other class is called derived class or child class. The derived class is a power packed class, as it can add additional attributes and methods and thus enhance its functionality.

Notes

The main advantage of inheritance is

- It represents real world relationships well
- It provides reusability of code
- It supports transitivity

16.3 Types of Inheritance

There are different types of inheritance viz., Single Inheritance, Multiple inheritance, Multilevel inheritance, hybrid inheritance and hierarchical inheritance.

1. Single Inheritance

When a derived class inherits only from one base class, it is known as single inheritance

2. Multiple Inheritance

When a derived class inherits from multiple base classes it is known as multiple inheritance

3. Hierarchical inheritance

When more than one derived classes are created from a single base class , it is known as Hierarchical inheritance.